Questions Are All You Need: A Question-Centric Framework for Automated Data Visualization

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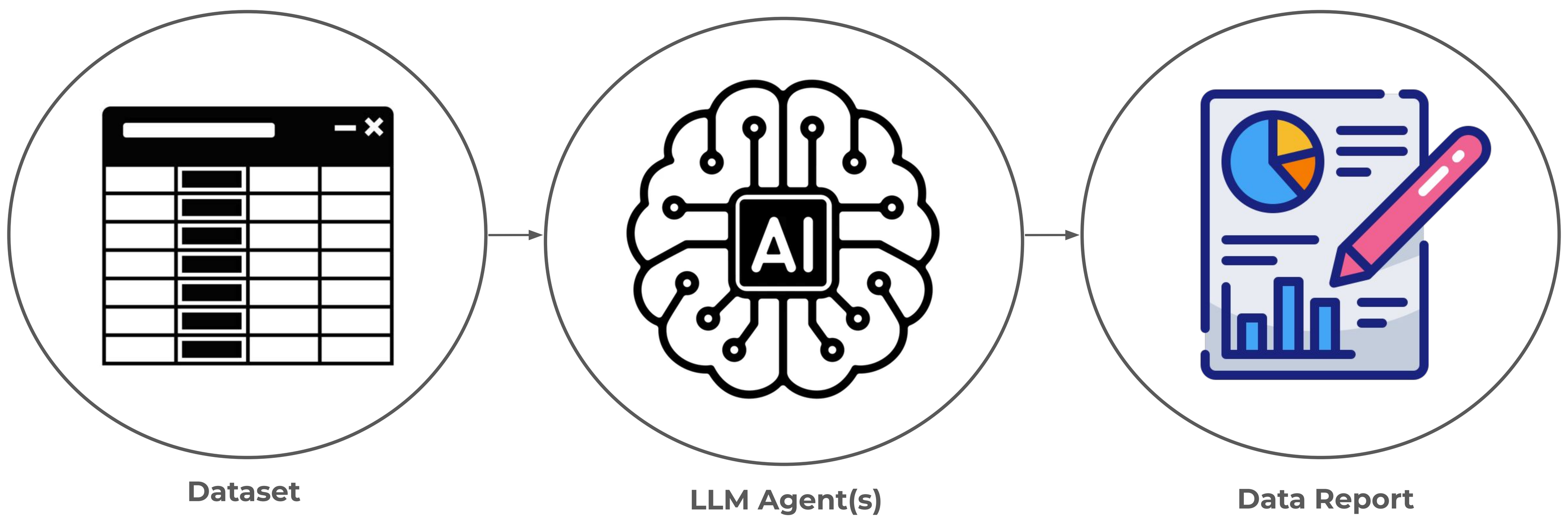
Problem Statement

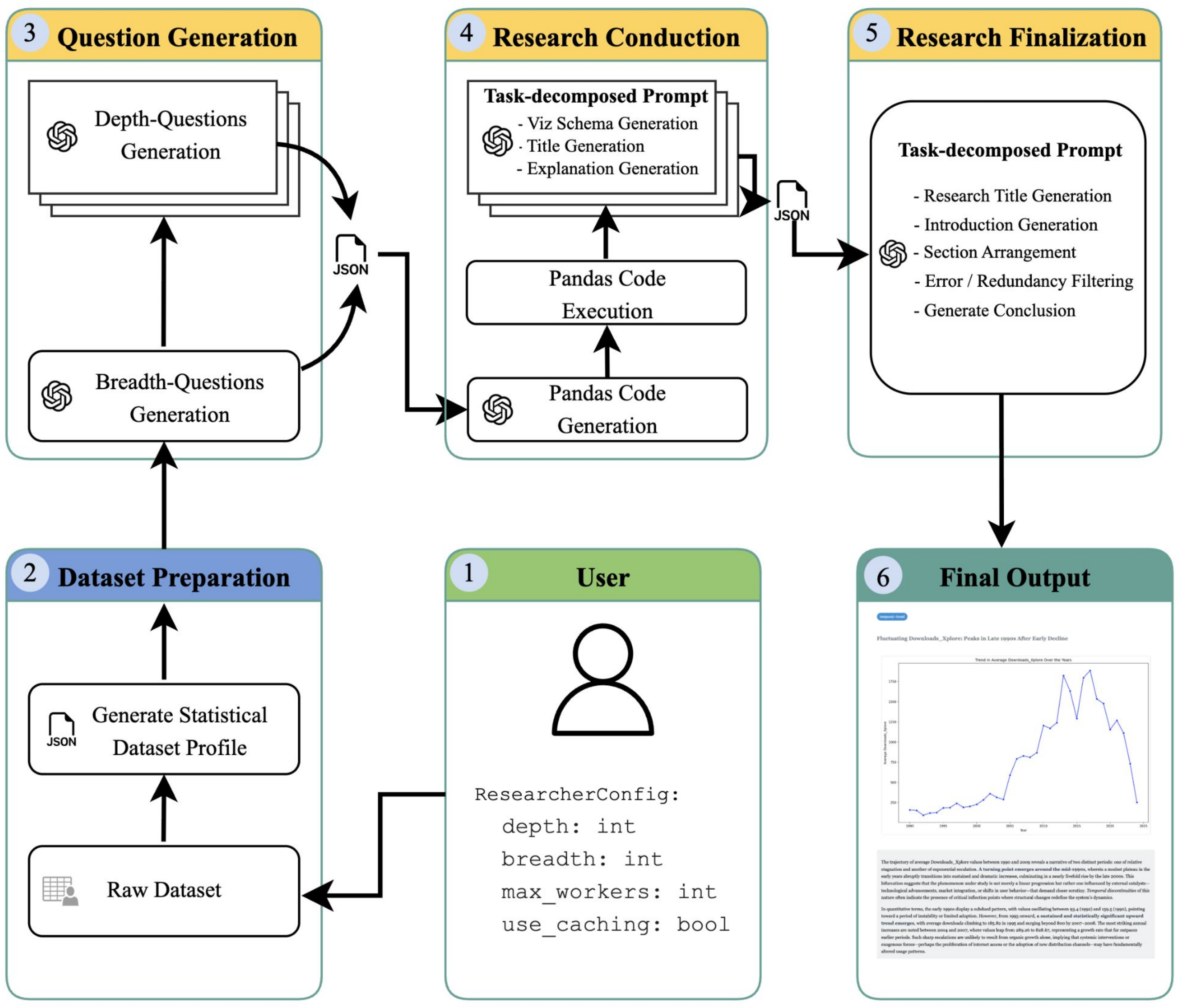


- Develop an agentic Large Language Model (LLM) workflow capable of **autonomously** analyzing arbitrary datasets and generating insightful, coherent, and visually compelling data reports.
- Ensure **robust generalization** across heterogeneous datasets by minimizing dependency on dataset-specific tuning or manual intervention.
- Produce **reliable** and **logically consistent** analytical outputs with effective visualizations that accurately reflect the underlying data.

Key Decisions

- **Question-First Reformulation**: Replaced traditional feature-engineering pipelines with a question-centric approach, where LLMs generate analytical questions from dataset profiles rather than raw data.
- **Pandas-First Computation Strategy**: Delegated all data processing to deterministic pandas execution, using the LLM only to generate code. This minimized token usage, improved scalability, and made computations reproducible and debuggable.
- **Concurrent Execution Architecture**: Implemented I/O-bound concurrency where each question executes independently in parallel threads—achieving high throughput and fault isolation without CPU-level parallelism.
- **Structured Prompt Decomposition**: Transitioned from verbose, example-heavy prompts to succinct, step-by-step task prompts with unified JSON outputs.
- **Abstraction through the `Researcher` Class**: Encapsulated all LLM operations, caching, and configuration logic into a modular abstraction (`Researcher` + `ResearchConfig`), enabling flexible experimentation, systematic parameter control, and reproducible analysis across datasets.

Architecture Overview



- **Stage 1: Question Generation**
  - Generates breadth questions sequentially to ensure comprehensive coverage, then **concurrently expands each breadth question into depth questions** using independent threads.
- **Stage 2: Research Conduction**
  - Processes all questions concurrently:
    - (1) LLM generates pandas code from question metadata and sample data
    - (2) code executes deterministically to compute results
    - (3) LLM receives question and computed data, returning visualization code, title, and explanation atomically.
- **Stage 3: Report Finalization**
  - Arranges results into a coherent report sequentially. The LLM receives all research results through a structured-output, multi-task prompt that decomposes report generation into explicit steps.

Experiment

Dataset	Rows	Cols	Time (s)	Vizs	Time/Viz (s)
Insurance	1,338	7	53.27	9	5.92
VisPub	3,877	20	140.89	9	15.65
Books	6,810	12	166.09	9	18.45
Movies	10,000	8	262.05	9	29.12

- **Consistent Performance Scaling:** The framework demonstrated predictable execution times across four datasets (1K–10K rows), with average completion around 2.5 minutes and stable scaling relative to dataset size and question complexity.
- **Zero-Shot Generalization:** Identical prompts successfully processed all datasets without modification, confirming strong **transferability** and **robustness** of the structured, question-centric design.

Conclusion & Future Works

- **Question-Centric Paradigm:** Reframing automated data analysis around *questions* instead of *features* establishes a more resilient and interpretable foundation for LLM-driven visualization
- **Scalable and Reproducible Workflow:** The modular architecture supports concurrent execution, caching, and structured prompting, ensuring consistent performance across diverse datasets.
- **Subjectivity of Analytical Quality:** While outputs were coherent and domain-appropriate, evaluating *insightfulness* and *narrative depth* remains inherently subjective—necessitating **human evaluation studies** for objective quality benchmarks.
- **Future Enhancements:** Incorporating **iterative question refinement**, **domain-specific vocabulary**, and **quantitative metrics** for narrative and visualization quality would further strengthen interpretability and analytical reliability.